

Introduction and Basic operations with vectors — Problem Set

Warm-up

1. Find the magnitude of the vector $\mathbf{v} = 3\mathbf{i} - 4\mathbf{j}$.
2. Given $\mathbf{a} = \begin{pmatrix} 2 \\ -5 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$, calculate $2\mathbf{a} + \mathbf{b}$.
3. State the position vector of the point $P(6, -8, 2)$.
4. Find the unit vector in the direction of $\mathbf{u} = \begin{pmatrix} 8 \\ 15 \end{pmatrix}$.
5. Write the vector equation of a line passing through $(0, 0)$ with direction vector $\mathbf{d} = \mathbf{i} + \mathbf{j}$.
6. If $\vec{OA} = \mathbf{a}$ and $\vec{OB} = \mathbf{b}$, express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$.
7. Determine the value of m if the vector $\begin{pmatrix} 2 \\ m \end{pmatrix}$ is parallel to $\begin{pmatrix} 6 \\ -9 \end{pmatrix}$.
8. Identify the center and radius of the circle given by $|\mathbf{r} - (3\mathbf{i} + 2\mathbf{j})| = 5$.
9. A line has the equation $\mathbf{r} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 4 \end{pmatrix}$. Find the coordinates of the point when $\lambda = 3$.
10. Simplify the vector expression $\vec{PQ} + \vec{QR} + \vec{RP}$.

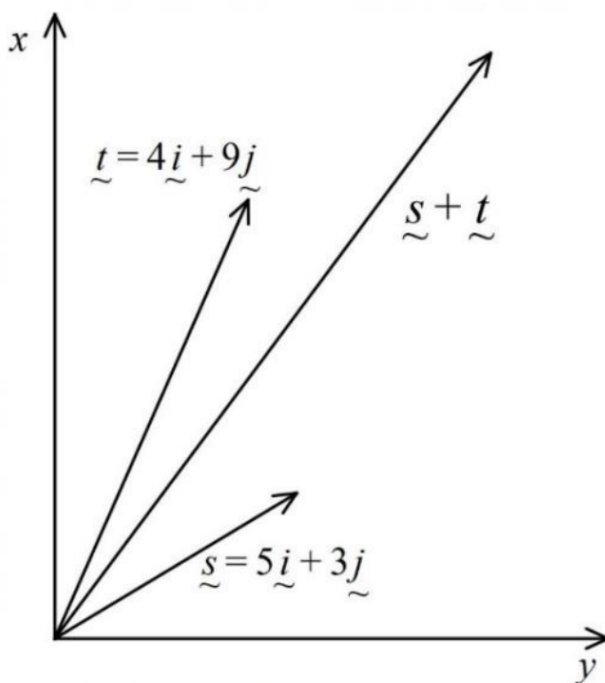
Skill-building

1. Two vectors are given by $\mathbf{u} = k\mathbf{i} + 3\mathbf{j}$ and $\mathbf{v} = 4\mathbf{i} - 2\mathbf{j}$. Find the value of k such that $\mathbf{u}$ is parallel to $\mathbf{v}$.
2. In triangle ABC , M is the midpoint of BC . If $\vec{AB} = \mathbf{c}$ and $\vec{AC} = \mathbf{b}$, express $\vec{AM}$ in terms of $\mathbf{b}$ and $\mathbf{c}$.
3. A sphere has the vector equation $|\mathbf{r} - (2\mathbf{i} - \mathbf{j} + 3\mathbf{k})| = \sqrt{14}$. Determine if the origin $(0, 0, 0)$ lies on the surface of this sphere.
4. Find the vector equation of the line passing through $A(1, 4, -2)$ and $B(3, 0, 5)$.
5. The line $\mathbf{r} = \begin{pmatrix} 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} 3 \\ k \end{pmatrix}$ passes through the point $(-4, 3)$. Find the value of k .
6. Convert the vector equation $\mathbf{r}(t) = (2 + 3\cos t)\mathbf{i} + (1 + 3\sin t)\mathbf{j}$ into a Cartesian equation.
7. Determine the point of intersection between the lines $\mathbf{r}_1 = \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{r}_2 = \begin{pmatrix} 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.
8. Given $\mathbf{a} = \mathbf{i} - 2\mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = 3\mathbf{i} + p\mathbf{j} - \mathbf{k}$, find the value of p if the vectors are perpendicular.

9. Points A, B, C are such that $\vec{AB} = 3\mathbf{i} - 2\mathbf{j}$ and $\vec{BC} = \mathbf{i} + 5\mathbf{j}$. Find the length of the vector $\vec{AC}$.
10. A particle moves such that its position is $\mathbf{r}(t) = t^2\mathbf{i} + (2t - 1)\mathbf{j}$. Find the Cartesian equation of its path by eliminating t .

Easier Exam Questions

1. (Sydney Girls 2025 Q1) Find the length of the vector $\begin{pmatrix} -7 \\ 8 \end{pmatrix}$.
- (A) 13
 (B) $\sqrt{15}$
 (C) $\sqrt{113}$
 (D) $\sqrt{117}$
2. (Sydney Tech 2023 Q3) Which of the following vectors is parallel to the vector $\vec{OQ} = 3\mathbf{i} - \mathbf{j}$?
- (A) $\vec{OA} = \mathbf{i} - 3\mathbf{j}$
 (B) $\vec{OB} = 12\mathbf{i} - 6\mathbf{j}$
 (C) $\vec{OC} = 3\mathbf{i} + \mathbf{j}$
 (D) $\vec{OD} = -12\mathbf{i} + 4\mathbf{j}$
3. (Cheltenham 2023 Q6) The diagram below shows the vectors $\mathbf{s}$ and $\mathbf{t}$ along with the vector $\mathbf{s} + \mathbf{t}$.



What is the value of $|\mathbf{s} + \mathbf{t}|$?

- (A) 12
- (B) 15
- (C) 16
- (D) 20

4. (Sydney Tech 2025 Q4) The vector $\overrightarrow{AB}$ joins the points $A(-5, 4)$ and $B(3, 10)$. Three other vectors are given:

$$\underline{u} = \underline{i} + 2\underline{j}$$

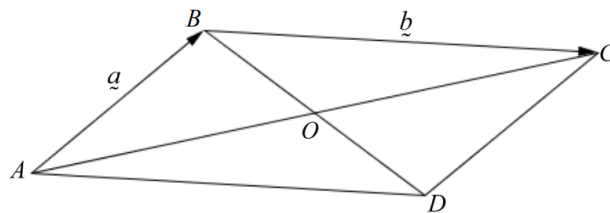
$$\underline{v} = 2\underline{i} + \underline{j}$$

$$\underline{w} = 3\underline{i} + \underline{j}$$

Which of the following vectors is parallel to $\overrightarrow{AB}$?

- (A) $\underline{u} - \underline{v}$
- (B) $\underline{u} + \underline{v}$
- (C) $\underline{w} + \underline{v}$
- (D) $\underline{w} + \underline{u}$

5. (Hurlstone 2021 Q6)



In the parallelogram $ABCD$ shown, the point of intersection of the diagonals is O .

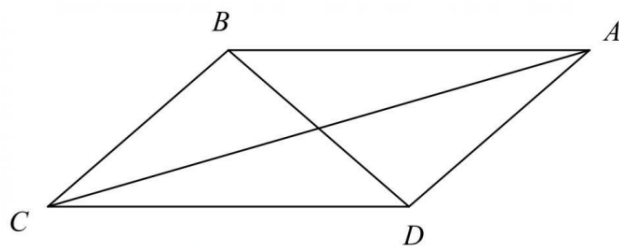
$$\vec{AB} = \underline{a} \text{ and } \vec{BC} = \underline{b}$$

What is the vector $\vec{OC}$ equal to?

- (A) $\frac{1}{2}(\underline{a} - \underline{b})$
- (B) $\frac{1}{2}(\underline{a} + \underline{b})$
- (C) $\frac{1}{2}\underline{a} - \underline{b}$
- (D) $\frac{1}{2}\underline{b} - \underline{a}$

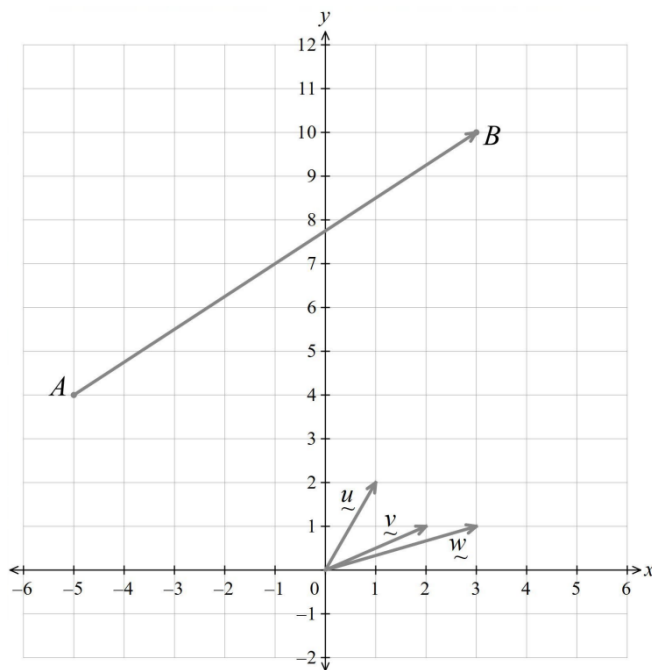
6. (Barker 2020 Q4) If $ABCD$ is a parallelogram, which vector is equal to $\overrightarrow{AB} + \overrightarrow{BD}$?

- (A) $\overrightarrow{DA}$
- (B) $\overrightarrow{BC}$
- (C) $\overrightarrow{AC}$
- (D) $\overrightarrow{CD}$



Harder Exam Questions

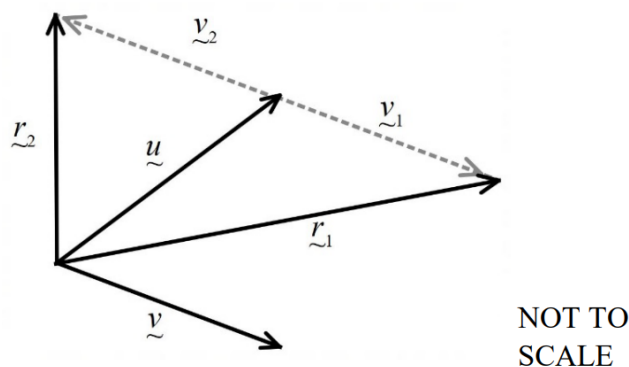
- (Fort St 2024 Q11a) Let P be the point $(12, 5)$. Find the position vector of point P , and hence find a unit vector in the direction of the position vector. **2**
- (Ascham 2024 Q11b) Given the vectors $\mathbf{a} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 3 \end{pmatrix}$, find $\mathbf{c}$ if $\mathbf{a} + 2\mathbf{b} + \mathbf{c} = \mathbf{0}$. **[2]**
- (Cheltenham Girls 2024 Q7) On the diagram below, the vector $\vec{AB}$ is shown joining $A(-5, 4)$ and $B(3, 10)$.



There are three other vectors shown on the diagram: $\vec{u} = \vec{i} + 2\vec{j}$ $\vec{v} = 2\vec{i} + \vec{j}$ $\vec{w} = 3\vec{i} + \vec{j}$

Which of the following vectors is parallel to $\vec{AB}$?

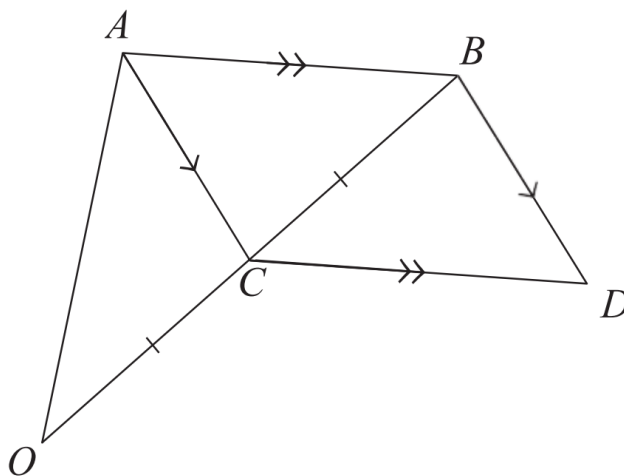
- $\vec{u} - \vec{v}$
 - $\vec{u} + \vec{v}$
 - $\vec{w} + \vec{v}$
 - $\vec{w} + \vec{u}$
- (Abbotsleigh 2024 Q6) The diagram shows two vectors $\vec{u}$ and $\vec{v}$. Two resultant vectors, $\vec{r}_1$ and $\vec{r}_2$, are constructed using $\vec{v}_1$ and $\vec{v}_2$ which are parallel to, and equal in length to $\vec{v}$. Which statement is true?



- (A) $\underline{r_1} = \underline{u} - \underline{v}$ and $\underline{r_2} = \underline{u} + \underline{v}$
 (B) $\underline{r_1} = \underline{v} + \underline{u}$ and $\underline{r_2} = \underline{v} - \underline{u}$
 (C) $\underline{r_1} = \underline{v} - \underline{u}$ and $\underline{r_2} = \underline{u} - \underline{v}$
 (D) $\underline{r_1} = \underline{u} + \underline{v}$ and $\underline{r_2} = \underline{u} - \underline{v}$

5. (Hornsby Girls 2020 Q7)

The position vectors of points A and B are $\underline{a}$ and $\underline{b}$ respectively. Point C is the midpoint of OB and point D is such that $ABDC$ is a parallelogram. Which of the following is the position



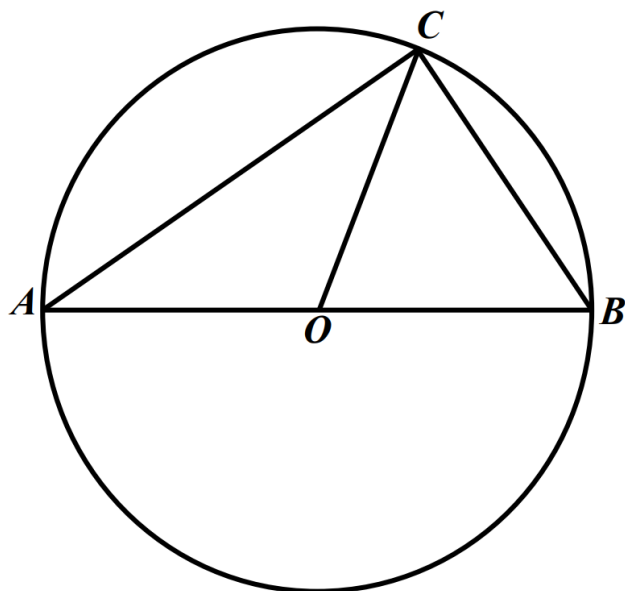
vector of D ?

- (A) $\frac{3}{2}\underline{b} + \underline{a}$
 (B) $\frac{3}{2}\underline{b} - \underline{a}$
 (C) $\frac{1}{2}\underline{b} - \frac{1}{2}\underline{a}$
 (D) $\frac{1}{2}\underline{b} - \underline{a}$

6. (Girraween 2022 Q9) In the diagram below, AOB is a diameter of the circle with centre O . C is a point on the circumference of the circle

If $\vec{OC} = \underline{r}$ and $\vec{BC} = \underline{s}$, then $\vec{AC}$ would be equal to which of the following vectors?

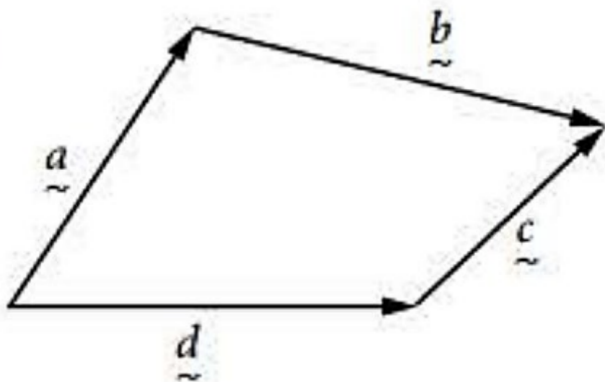
- (A) $\underline{r} + 2\underline{s}$



- (B) $\tilde{r} - 2\tilde{s}$
- (C) $2\tilde{r} + \tilde{s}$
- (D) $2\tilde{r} - \tilde{s}$

7. (St George Girls 2021 Q6) The points P, Q and R lie on a straight line. The vector $\overrightarrow{PQ} = \begin{pmatrix} -4 \\ 6 \end{pmatrix}$ and the vector $\overrightarrow{PR} = 2\overrightarrow{PQ}$. The coordinates of P are $(2, -2)$. What are the coordinates of R ?
- (A) $(-2, 4)$.
 - (B) $(-6, 8)$
 - (C) $(-6, 10)$
 - (D) $(6, -10)$

8. (Hurlstone 2025 Q6) Four vectors, $\tilde{a}$, $\tilde{b}$, $\tilde{c}$ and $\tilde{d}$ are shown in the diagram. Which one of the



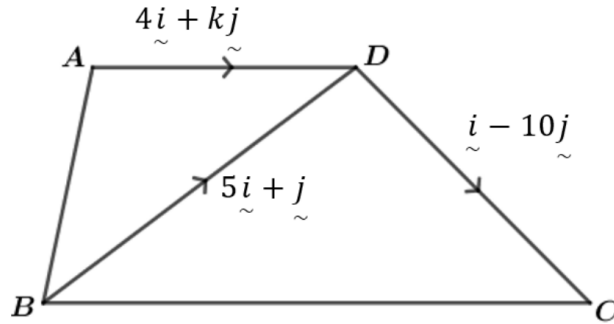
following statements is true?

- (A) $\tilde{a} + \tilde{c} = \tilde{b} + \tilde{d}$
- (B) $\tilde{a} + \tilde{b} = \tilde{c} + \tilde{d}$

(C) $\underline{\underline{a}} + \underline{\underline{b}} + \underline{\underline{c}} + \underline{\underline{d}} = 0$

(D) $\underline{\underline{b}} + \underline{\underline{c}} = \underline{\underline{a}} + \underline{\underline{d}}$

9. (Hurlstone 2023 Q12a) The figure below shows a trapezium $ABCD$, where AD is parallel to BC . The following information is given for this trapezium:



$\overrightarrow{BD} = 5\mathbf{i} + \mathbf{j}$, $\overrightarrow{DC} = \mathbf{i} - 10\mathbf{j}$ and $\overrightarrow{AD} = 4\mathbf{i} + k\mathbf{j}$, where k is an integer.

- (i) Use vector algebra to find the value of k . **2**

- (ii) Find the length of $\overrightarrow{AB}$. **1**

10. (Moriah 2023 Q6) A , B and C are collinear points with position vectors $\underline{a}$, $\underline{b}$ and $\underline{c}$ respectively. B lies between A and C . Given that $|\overrightarrow{BC}| = \frac{1}{2}|\overrightarrow{AB}|$, which of the following expressions is equal to $\underline{c}$?

(A) $\frac{1}{2}\underline{a} - \frac{3}{2}\underline{b}$

(B) $\frac{3}{2}\underline{a} - \frac{1}{2}\underline{b}$

(C) $\frac{3}{2}\underline{b} - \frac{1}{2}\underline{a}$

(D) $\frac{3}{2}\underline{b} - \frac{3}{2}\underline{a}$